

Final Examination – PROG8111-25S-Sec1

Total Marks: 25, Time: 3 Hours

Please read over the entire exam carefully before starting!

Objective:

Build a React Native mobile application using TypeScript that functions as a basic **Student Portal**. The app should implement drawer navigation, manage state using Zustand, display data with FlatList, and store personal notes using a local SQLite database. The UI should be clean, styled, and interactive.

You must use **React Native CLI with TypeScript** and should adhere to class learnings of this course. Various screens of the application should look like the images provided at the end of this document.

The ReactNative Application should be named **FinalExam_StudentId**. Replace StudentId with your college id.

Features to Implement:

1. Simulated Course Data

- Create a **courses.ts** file with a static array of at least 10 course objects.
- Each course object should contain: courseId, courseName, instructor, credits, semester.

2. Drawer Navigation

- Implement a **Drawer Navigator** with the following screens:
 - **Courses**
 - **Favorites**
 - **Notes**
 - **About**

3. Courses Screen with FlatList

- Display all courses from courses.ts using FlatList.
- Each item must show: courseId, courseName, instructor.
- Add a button to toggle “Add to Favorites” / “Remove from Favorites”.
- On tapping a course, open a **modal** to show full details:
 - courseId, courseName, instructor, credits, semester.

4. Favorites Screen

- Use Zustand to store and manage a global list of favorite courses.
- Show all favorited courses using a FlatList.

5. Notes Screen with SQLite

- Allow the user to:
 - Add a note (with title and content)
 - View all saved notes
 - Delete individual notes
- Use SQLite to store notes locally on the device.

6. About Screen

- Add a simple **Settings** screen with static content like brief introduction to app, and developer details

Styling:

The user interface (UI) of the Student Portal application must maintain a clean, modern, and visually consistent design across all screens. Components such as buttons, cards, and modals should be styled with appropriate spacing, colors, and typography to enhance readability and user experience.

Additional Requirements:

- Use functional components and React Hooks.
- Maintain proper code structure: separate folders for screens, state (Zustand), navigation, and data.
- Add meaningful comments to explain important logic.
- Input validation where appropriate (e.g., note fields shouldn't be empty).

Submission Requirements:

Upload the following items into the submission folder titled 'Final Examination'

1. Release APK
2. A Word document with all relevant screenshots of your running application and the entire code commented meaningfully. Code pasted as image will not be considered for

evaluation. **Do Not zip this document.** This Word document should be named as **StudentId.docx**. Replace StudentId with your college id.

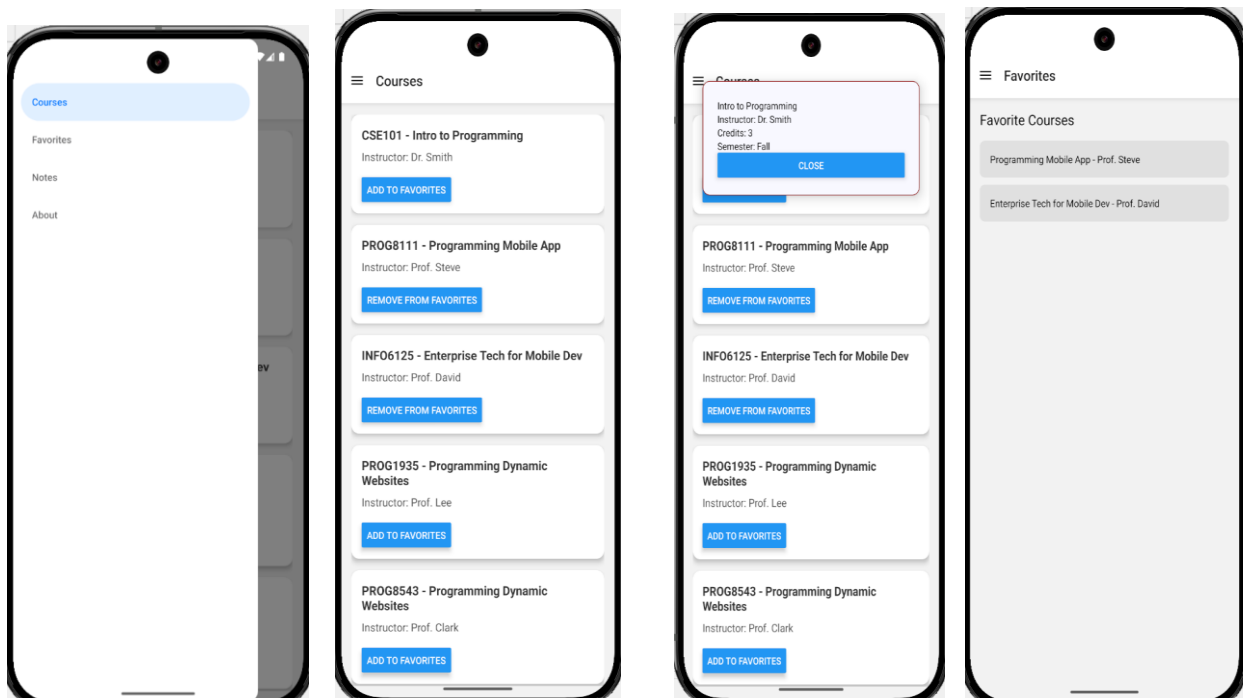
3. Submit a zipped version of your application folder, ensuring that the **node_modules** and android **build** directories **are deleted** beforehand (as explained in the class).

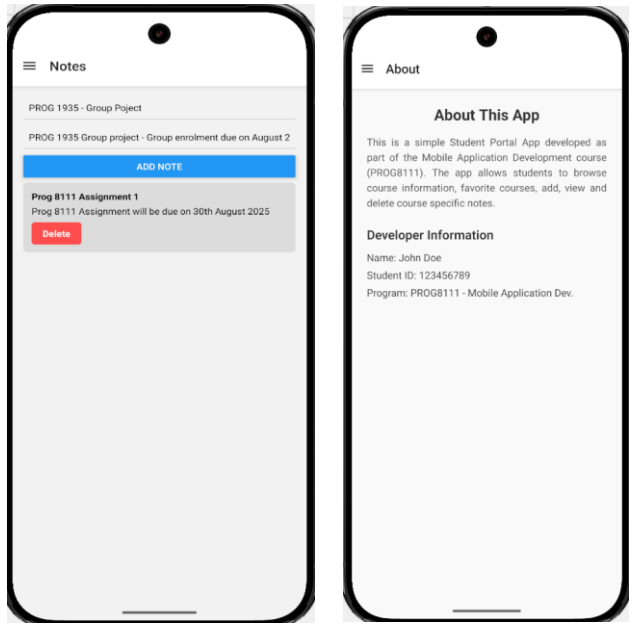
Submissions containing these folders will not be accepted or evaluated under any circumstances, and will be awarded a grade of 0 marks.

Important Note:

- If a package installation fails for any valid reason, you must include a clear screenshot in the Word document showing both the command you entered and the exact error message. Screenshots that do not clearly display both the command and the error will not be accepted.
- The submission will not be considered for evaluation without the zip of the React Native Application, Word document and the release APK.
- There will be no late penalty scheme available for final examination. Late submissions will not be considered for evaluation under any circumstances.

Images of Screens:





Deductions	
Word document not submitted with commented code	100%
Word document doesn't include screenshots	25%
APK Missing	50%
Application Zip Missing	100%
Run time errors	15% for each error